9	**(a)**	State what is meant by *simple harmonic motion.* ....[2]

Solution:

A periodic motion in which the body's acceleration is directly proportional to its displacement from its equilibrium position — 1

its acceleration is always in opposite direction to its displacement — 1

A long strip of springy steel is clamped at one end so that the strip is vertical. A mass *m* of 65 g is attached to the free end of the strip, as shown in Fig. 9.1.

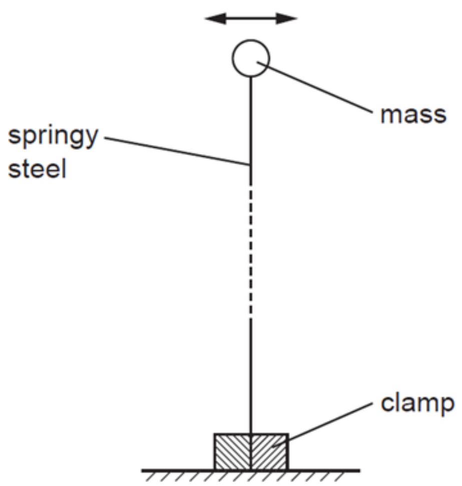


Fig. 9.1

The mass is pulled to one side and then released. The variation with time *t* of the horizontal displacement of the mass is shown in Fig. 9.2.

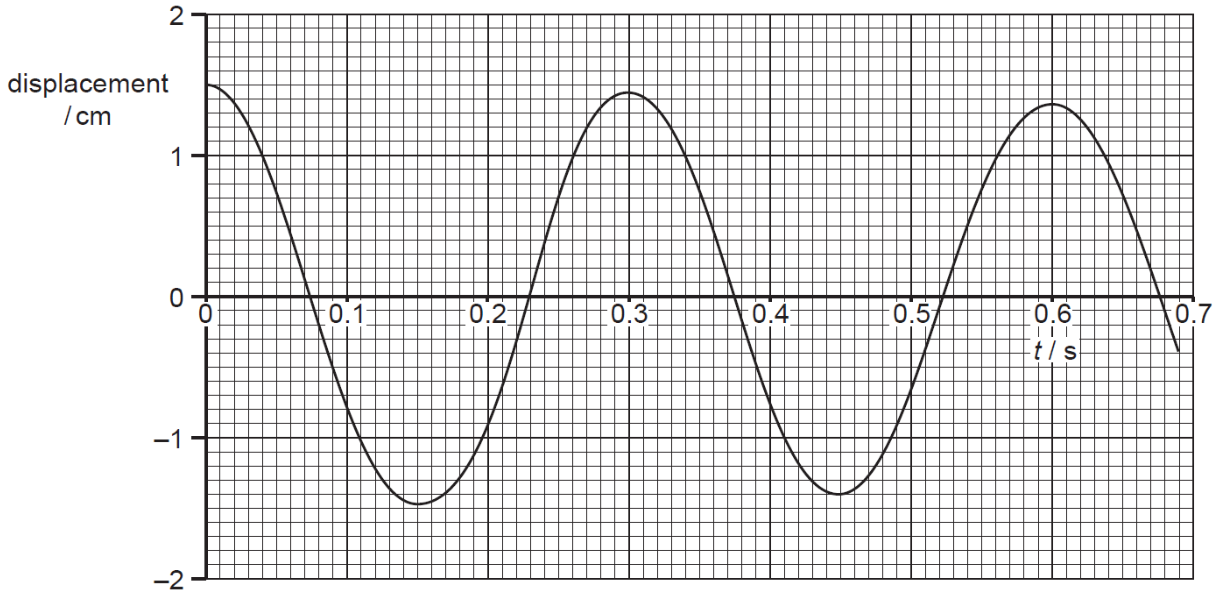


Fig. 9.2

The mass undergoes damped oscillation.

(b) **(i)** Suggest, with a reason, whether the damping is light, critical or heavy.

....

....

.... [2]

Solution:

The amplitude of the oscillation is decreasing gradually / oscillations continue for a long time — 1

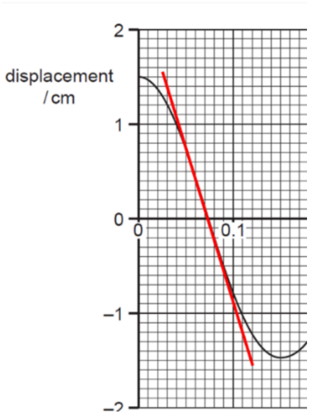
Light damping — 1

(ii) **1.** Using the data from Fig. 9.2, determine the maximum speed of the oscillation. State one time at which it is moving at maximum speed.

maximum speed =..........................m s^{-1}

time = ... s [2]

Solution:



speed

= gradient of the tangent

$= \left| \dfrac{(-1.2-1.4)}{(0.11-0.03)} \right|$

= 32.5 cm s^{-1} = 0.325 m s^{-1} 1

Time = 0.075 s, 0.23 s, 0.375 s, 0.52 s, 0.675 s 1

2. Sketch a graph showing the variation of the velocity of the mass *v* with its displacement from the equilibrium position in Fig. 9.3 for the first complete period of the oscillation.

Label the axes with an appropriate scale.

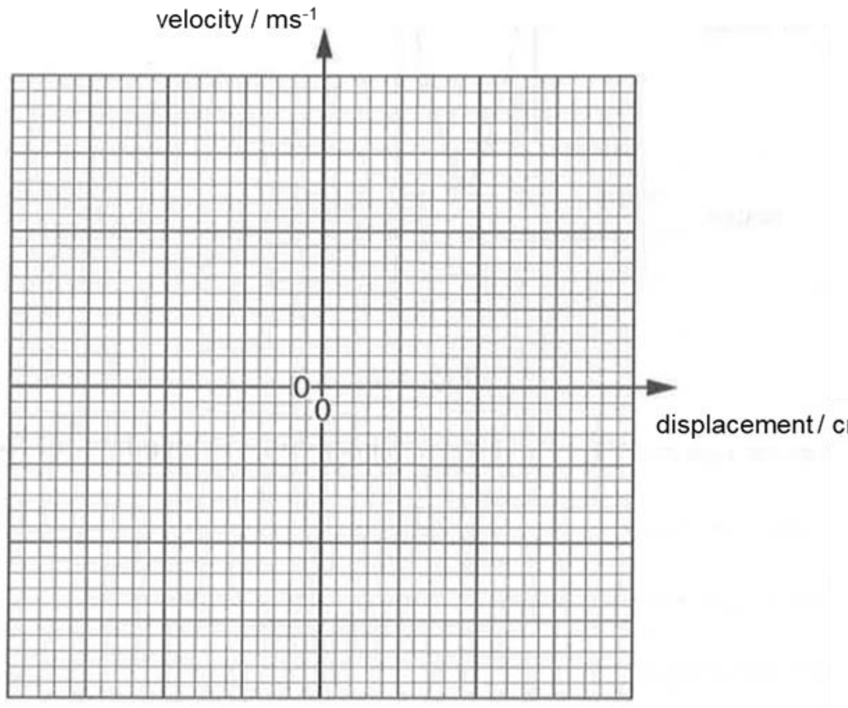
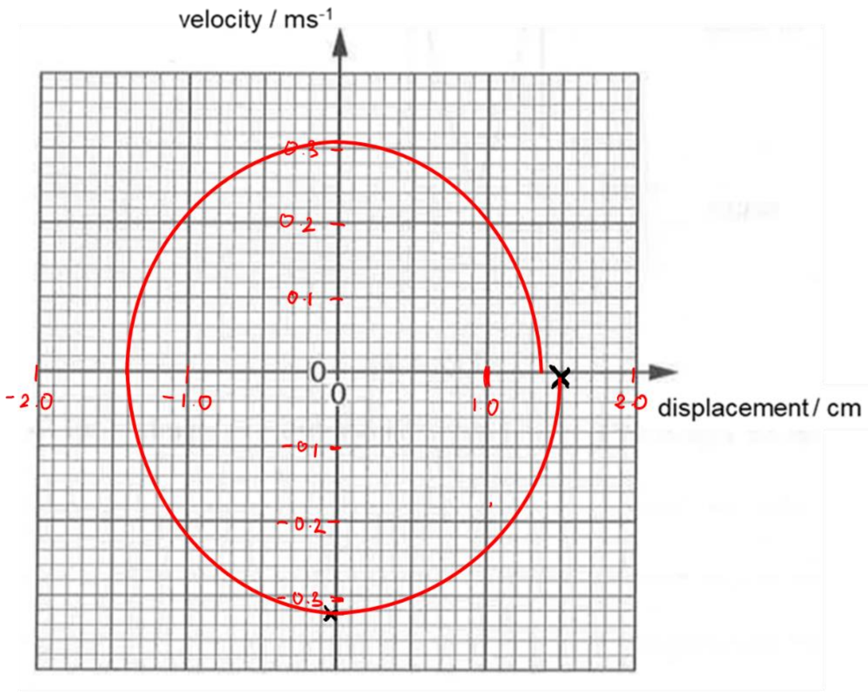


Fig. 9.3

[3]

		L3		**Solution:**



1 mark – graph begins at (1.5 cm, 0)
1 mark - graph shape (clockwise, ellipse, not "closed")
1 mark – correct intercepts of axes (v and x values smaller than the previous intercept)

(iii) After eight complete oscillations of the mass, the amplitude of vibration is reduced from 1.5 cm to 1.1 cm.

1. Show that the total energy of a body oscillating in simple harmonic motion is given by

$$E_T = \frac{1}{2} m\omega^2 x_0^2$$

where ω is the angular frequency of the oscillation and x_0 is its amplitude

[3]

				Solution: $v = \pm\omega\sqrt{x_0^2 - x^2}$ speed is maximum when $x = 0$ $v_0 = \pm\omega\sqrt{x_0^2} = \omega x_0$ The total energy of the oscillation = maximum kinetic energy $= \frac{1}{2}mv_0^2$ $= \frac{1}{2}m\omega^2 x_0^2$	1 1 1
			2.	Calculate the angular frequency of the oscillations. angular frequency = ……………………………rad s^{-1} [2]	
				Solution: $\omega = 2\pi/T$ $= 2\pi/0.3$ $= 20.9$ rad s^{-1}	1 1
			3.	Calculate the loss of energy after eight complete oscillations. loss of energy = ……………………………J [2]	
				Solution: Loss of Energy = total initial energy of oscillation – total final energy of oscillation $= \frac{1}{2} m\omega^2 x_{o,initial}^2 - \frac{1}{2} m\omega^2 x_{o,final}^2$ $= \frac{1}{2} \times 0.065 \times 20.9^2 \times (0.015^2 - 0.011^2)$ = 0.00147 = 0.0015 J	1 1
			4.	Suggest with a reason, whether after another eight complete oscillations, the amplitude will be 0.7 cm.	

				....
				....
				.... [2]

Solution:

amplitude reduces exponentially / does not decrease linearly 1
so will be not be 0.7 cm 1

(c) The variation of kinetic energy E_k of the mass with displacement x from its equilibrium position for the first period of oscillation is shown in Fig. 9.4.

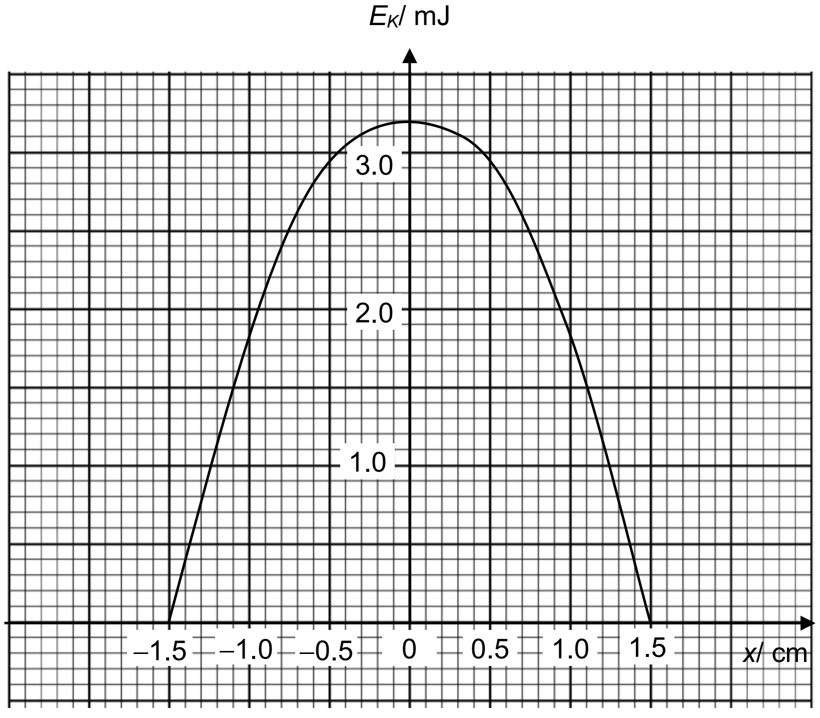


Fig. 9.4

The mass loses energy such that, after some time, its maximum kinetic energy is reduced by 1.5 mJ.

Use Fig. 9.4, without further calculation, to determine the amplitude of the oscillations. Show your construction on Fig. 9.4.

		amplitude = ... cm [2] [Total: 20]	
		Solution: When the objects loses KE, the entire graph will be translated downwards by 1.5 mJ. The point that will now have 0 KE, or the amplitude will be the displacement at which the E was originally 1.5 mJ Construction of horizontal line at 1.5 mJ or correctly translated KE graph amplitude = 1.1 cm If shifted curve is shown with the correct 3 key points (i.e. max and the two amplitude points, 2 marks). If shifted curve is shown with only the correct max point (wrong amplitude values), 1 mark	 1 1